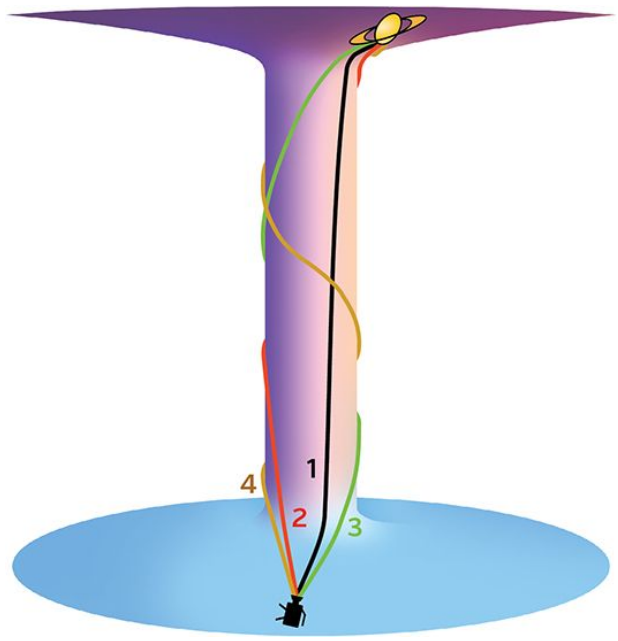


Fig. 15.3. Light rays from Saturn, through the wormhole, to the camera.

Can you explain the fifth and sixth images? and explain why the images shrink when the wormhole is lengthened? and explain why the images appear to emerge from the edge of the wormhole's crystal-ball mouth and move inward?

The Wormhole's Lensing Width

Having understood how the wormhole's length affects what the camera sees, we then fixed the length to be fairly short, the same as the wormhole's radius, and varied the gravitational lensing. We increased the wormhole's lensing width from near zero to about half the wormhole's diameter and computed what that did to the images the camera sees. Figure 15.4 shows the two extremes.

With very small lensing width, the wormhole shape (upper left) has a sharp transition from the external universe (horizontal sheets) to the wormhole throat (vertical cylinder). As seen by the camera (upper